More Hops, More Graph: Query Complexity Decides How Much to Trust the Cross-Encoder in Graph-Augmented Re-ranking

Tanish Verma **Narmety Vamshi** **Pruthviraj Sharad Salunke**
Mehta Family School of Data Science & Artificial Intelligence
Indian Institute of Technology Roorkee
{tanish_v, vamshi_n, pruthviraj_ss}@mfs.iitr.ac.in

Abstract

Graph-augmented rerankers such as ReGAT-BERT fuse a cross-encoder score s_{sem} and a graph-attention contextual score s_{ctx} with one global weight, $\alpha\, s_{sem} + (1-\alpha)\, s_{ctx}$. We hypothesise that this weight should depend on the query: simple single-hop queries should rely on the cross-encoder ($\alpha > 0.5$) and complex multi-hop queries on the contextual score ($\alpha < 0.5$). We test this in a controlled ablation (four score normalisers $\times$ five ways of setting α, all upstream models frozen) on MS MARCO and four multi-hop collections. The results support the hypothesis for questions that genuinely require composing several passages. The best fixed α is 0.70–0.95 on MS MARCO but 0.40 on MuSiQue and 0.45 on 2WikiMultiHopQA, whose questions need more than two supporting passages. Within MuSiQue the optimal α *decreases monotonically with the number of hops* under every scale-free normaliser (test, min–max: 0.80, 0.55, 0.40 for 2, 3, 4 hops; validation: 0.40, 0.40, 0.25), and 4-hop questions favour the contextual score in all 14 normaliser/graph/split settings; per question, the preferred α is negatively correlated with the hop count ($p < 0.005$ on both splits). The mechanism is that the cross-encoder loses 12.5 nDCG@10 points from 2 to 4 hops while the GAT loses only 4.7 and overtakes it. On 2WikiMultiHopQA, comparison questions favour the cross-encoder and bridge-comparison questions the GAT ($\alpha = 0.20$–0.25); a query-only gate that never sees the type label gives comparison questions the highest weight on the cross-encoder under all three normalisers and gains significantly on bridge-comparison questions (up to +0.81 nDCG@10). Query-dependent weighting pays off where it exploits this structure: a type-aware α significantly improves three multi-hop collections, and 7 of the 11 significant gains of learned gates are on multi-hop data (up to +1.37). The effect is absent on HotpotQA, where the cross-encoder dominates.

1 Introduction

Neural rerankers increasingly combine a pointwise relevance model with evidence drawn from the other candidates: graph-based score regularisation (Diaz, 2005; Kurland and Lee, 2005), corpus-graph reranking (MacAvaney et al., 2022b), and graph-attention models over passages and their links (Asai et al., 2020). ReGAT-BERT (reg, n.d.) is one such design. A BERT cross-encoder (Nogueira and Cho, 2019) gives a semantic score $s_{sem}(q,p)$; a GAT (Veličković et al., 2018) refines bi-encoder passage embeddings over a graph of candidates, and the cosine with the query gives a contextual score $s_{ctx}(q,p)$. The final score is

$$s(q,p) = \alpha\, s_{sem}(q,p) + (1-\alpha)\, s_{ctx}(q,p), \quad (1)$$

with a single α for all queries.

A single weight is a strong assumption. A single-hop query ("*what is the capital of Peru*") can be answered by a passage judged in isolation, which is exactly what a cross-encoder does well. A multi-hop question ("*who directed the film whose lead actress was born in Lima*") needs passages that are relevant only through their relation to other passages, which is what a graph over the candidate set can encode. This motivates our hypothesis:

> **Hypothesis.** For simple single-hop queries $\alpha > 0.5$ (the semantic score matters more); for complex multi-hop queries $\alpha < 0.5$ (the contextual score matters more). Hence α should depend on the query, and an α conditioned on the query (Q), on the query and the score distributions (QS) or on the score distributions (S) should beat a fixed one.

We test the hypothesis in a controlled ablation in which only the fusion step changes (§3): candi-

dates, cross-encoder, bi-encoder and GAT are computed once and frozen, so every fusion variant sees identical inputs. MuSiQue (Trivedi et al., 2022), whose questions are annotated with 2, 3 or 4 hops, gives the most direct test. Our findings:

- **Multi-hop data moves the weight towards context.** The best fixed α is above 0.5 on MS MARCO and below 0.5 on MuSiQue (validation) and 2WikiMultiHopQA (§4.1).
- **More hops, lower α.** On MuSiQue the optimal α falls monotonically from 2 to 4 hops under every scale-free normaliser and, on test, for every graph we tried, and is below 0.5 for 4-hop questions in every setting (§4.2, Figure 2).
- **Why.** The cross-encoder, which reads each passage in isolation, degrades much faster with the number of hops than the graph-contextual score; across all question groups the optimal α tracks this reliability gap ($\rho = 0.79$; §4.3).
- **Question type matters as predicted on 2Wiki**, and a learned query-only gate recovers the type ordering without labels (§4.4).
- **Query-dependent α beats a fixed one where it exploits this structure**: type-aware weights and learned gates give their significant gains mostly on multi-hop data (§4.5).

We also report where the effect does not appear (§4.7): on HotpotQA, whose questions often admit single-hop shortcuts (Min et al., 2019), the cross-encoder keeps the weight.

2 Related Work

Score fusion and normalisation. Linear combination of retrieval scores goes back to CombSUM and its variants (Fox and Shaw, 1994); because scores live on different scales, per-query normalisation (min–max, z-score) is standard (Montague and Aslam, 2001), and rank-based fusion such as RRF (Cormack et al., 2009) avoids scores altogether. We compare four normalisers because the numerical value of α is only meaningful relative to the scales of s_{sem} and s_{ctx}.

Graph-based and contextual reranking. Score regularisation smooths scores over a document-similarity graph (Diaz, 2005); structural reranking uses graphs induced by language models (Kurland and Lee, 2005); adaptive reranking expands the candidate pool through a corpus graph (MacAvaney et al., 2022b). ReGAT-BERT (reg, n.d.) learns the propagation with a GAT and fuses it with a cross-encoder through a fixed weight, the component we study. We use GATv2 layers (Brody et al., 2022) in PyTorch Geometric (Fey and Lenssen, 2019).

Multi-hop retrieval. Multi-hop QA datasets (Yang et al., 2018; Ho et al., 2020; Trivedi et al., 2022) require combining several supporting paragraphs; retrievers for them follow hyperlinks or retrieve iteratively (Asai et al., 2020; Xiong et al., 2021). Many HotpotQA questions can be answered with single-hop shortcuts (Min et al., 2019), which motivated MuSiQue's bottom-up construction from single-hop questions so that every hop is necessary (Trivedi et al., 2022). This difference between the two datasets turns out to matter for fusion (§4.7).

3 Experimental Setup

3.1 Frozen pipeline

Candidates. MS MARCO and BEIR-HotpotQA use BM25 (Pyserini, k_1=0.9, b=0.4; Robertson and Zaragoza, 2009; Lin et al., 2021) top-100. MuSiQue, HotpotQA (distractor) and 2Wiki are closed pools of 20 or 10 paragraphs that always contain the gold paragraphs; we order each pool by in-pool BM25, which is also the tie-breaker everywhere. **Semantic score.** s_{sem} is the raw logit of `cross-encoder/ms-marco-MiniLM-L-6-v2` (Wang et al., 2020), never fine-tuned; on multi-hop data the passage is "*title*. *text*". **Contextual score.** Node features are frozen `msmarco-MiniLM-L6-cos-v5` embeddings (Reimers and Gurevych, 2019). For each query we build a graph over its candidates only (the query is not a node): a symmetrised k-NN graph on cosine similarity (k=8 for K=100, k=3 for closed pools, minimum cosine 0.3) with self-loops. Two GATv2 layers (4×128 heads, then 384) with a residual connection and LayerNorm refine the embeddings, and s_{ctx} is the cosine between the query embedding and the refined passage embedding. One GAT per dataset is trained once on a disjoint *GAT-train* split with a multi-positive listwise InfoNCE loss (τ=0.05; van den Oord et al., 2018), early-stopped on validation, and frozen. On every dataset GAT-only beats the raw bi-encoder on validation (e.g. MuSiQue 75.2 vs. 70.1 nDCG@10).

3.2 Fusion variants

Normalisers (per query over its candidates): *none*, *min–max*, *z-score*, *rank*. The primary normaliser,

fixed on MS MARCO validation before any multi-hop run, is min–max. With raw scores (*none*) α only reflects the ratio of the score scales (within a query, cross-encoder logits span about 12 units, cosines about 0.5), so we read α only under the three scale-free normalisers. **What α depends on.** FIXED: a grid search $\alpha \in \{0, 0.05, \ldots, 1\}$ on validation (main baseline, α_0) and a learned scalar. Q: $\alpha = \sigma(\mathrm{MLP}(\mathbf{q}))$ on the query embedding. S: $\alpha = \sigma(\mathrm{MLP}(\mathbf{f}))$ on 12 statistics of the raw score lists (mean, std, max, top-1/top-2 gap and softmax entropy of each score; Spearman correlation and top-m overlap between them). QS: both inputs. QS+C: QS plus query length and spaCy entity count. Each MLP has one hidden layer of 64 units, its output bias is initialised to $\mathrm{logit}(\alpha_0)$ and its last-layer weights near zero, so every gate starts as the fixed baseline. **Training.** Gates are trained on a disjoint *fusion-train* split with a multi-positive listwise softmax loss (Cao et al., 2007) (learned temperature) plus $\lambda(\alpha - \alpha_0)^2$, λ=0.01 (λ=0 as an ablation), AdamW (Loshchilov and Hutter, 2019), early stopping on validation nDCG@10, 5 seeds. **ReGAT-BERT-style row**: $a\, s_{sem} + b\, s_{ctx}$ with free a, b trained with pointwise BCE. **Oracle**: the per-query grid α that maximises that query's nDCG@10; an upper bound.

3.3 Data and complexity labels

Table 1 lists the collections. MS MARCO (Nguyen et al., 2016) is the single-hop reference, used under two protocols: **A** trains on MS MARCO train and tests on dev/small; **B** splits dev/small itself. Complexity labels, used only for analysis and never as gate inputs: MuSiQue hop count (2–4, plus the six composition types 2hop, 3hop1, 3hop2, 4hop1–3); 2Wiki question type (Ho et al., 2020): comparison, compositional, inference, bridge-comparison; HotpotQA and BEIR-HotpotQA (Thakur et al., 2021): bridge vs. comparison.

3.4 Evaluation

We report nDCG, MRR, Precision and Recall at $k \in \{1, 3, 5, 10, 50\}$, MAP and AllRecall@$k$ with ir_measures (MacAvaney et al., 2022a); nDCG@10 is the primary metric (full tables in Appendix A). Every choice (α_0, early stopping, the primary normaliser) is made on validation data. To study how the optimal weight depends on complexity independently of whether a gate learns it, we also compute, for each group of questions (e.g. all 4-hop questions), the fixed α that maximises the group's mean nDCG@10, with a bootstrap CI over the group's questions (Efron and Tibshirani, 1993), and a tie-robust per-question preference: for every question whose nDCG@10 changes with α, the centre of its optimal-α interval. These analyses are reported on both validation and test.

4 Results

4.1 Multi-hop data shifts the weight towards context

Table 2 and Figure 1 show the best fixed α per collection. On the single-hop reference the weight goes to the cross-encoder: $\alpha^* = 0.70$ (A) and 0.85 (B) on validation, 0.95 and 0.90 on test, under min–max and identically under z-score. On the two collections whose questions need more than two supporting passages on average (2.35 for MuSiQue, 2.43 for 2Wiki, vs. 1.06 for MS MARCO) it moves to the contextual score: $\alpha^* = 0.40$ on MuSiQue and 0.45 on 2Wiki (min–max validation; 0.45 and 0.50 under z-score, 0.45 and 0.55 under rank). 2Wiki is also the collection on which GAT-only matches CE-only (91.0 vs. 90.8 nDCG@10 on test). The overall MuSiQue test optimum (0.65) is higher than its validation optimum because the validation questions are drawn from the training set, like the GAT's training data, so the GAT is relatively stronger on them (CE$-$GAT -0.7 points on validation vs. $+4.3$ on test); the hop-count trend of the next section holds on both splits.

4.2 More hops, lower α

MuSiQue is the cleanest test of the hypothesis: its questions are built by composing 2, 3 or 4 single-hop questions so that every hop is needed (Trivedi et al., 2022), and the hop count is annotated. Figure 2 and Table 3 show that **the optimal weight on the cross-encoder falls as the number of hops grows**:

- **Test, min–max:** 0.80 [95% CI 0.65, 0.80] for 2 hops, 0.55 [0.55, 0.85] for 3 hops and 0.40 [0.25, 0.55] for 4 hops. **Validation:** 0.40, 0.40 and 0.25, i.e. every hop count favours the contextual score.
- **Every scale-free normaliser** shows the same monotone decline: z-score 0.80, 0.65, 0.40 (test) and 0.50, 0.40, 0.30 (validation); rank 0.95, 0.85, 0.00 (test) and 0.45, 0.30, 0.00 (validation).
- **Every graph**: retraining the GAT on an empty,

Dataset	Setting	K	GAT train	Fusion train	Val	Test	Gold/pool	Complexity labels
MS MARCO (A)	open (BM25)	100	30,000	10,000	2,000	6,980	–	proxies only
MS MARCO (B)	open (BM25)	100	1,870	1,150	698	1,745	–	proxies only
MuSiQue	closed pool	20	10,000	7,000	2,938	2,417	2.65	hops 2/3/4
HotpotQA	closed pool	10	20,000	10,000	2,000	7,405	2.00	bridge/comparison
BEIR-HotpotQA	open (BM25)	100	20,000	10,000	2,000	7,405	1.59	bridge/comparison
2Wiki	closed pool	10	20,000	10,000	2,000	12,576	2.44	4 types

Table 1: Datasets and splits (queries). Val is always carved from the training data; test is the official dev set (MS MARCO A: dev/small; MuSiQue, HotpotQA, 2Wiki: dev; BEIR: test). MS MARCO B splits dev/small itself, so the cross-encoder never saw any of its queries. Gold/pool = mean number of relevant passages among the K test candidates.

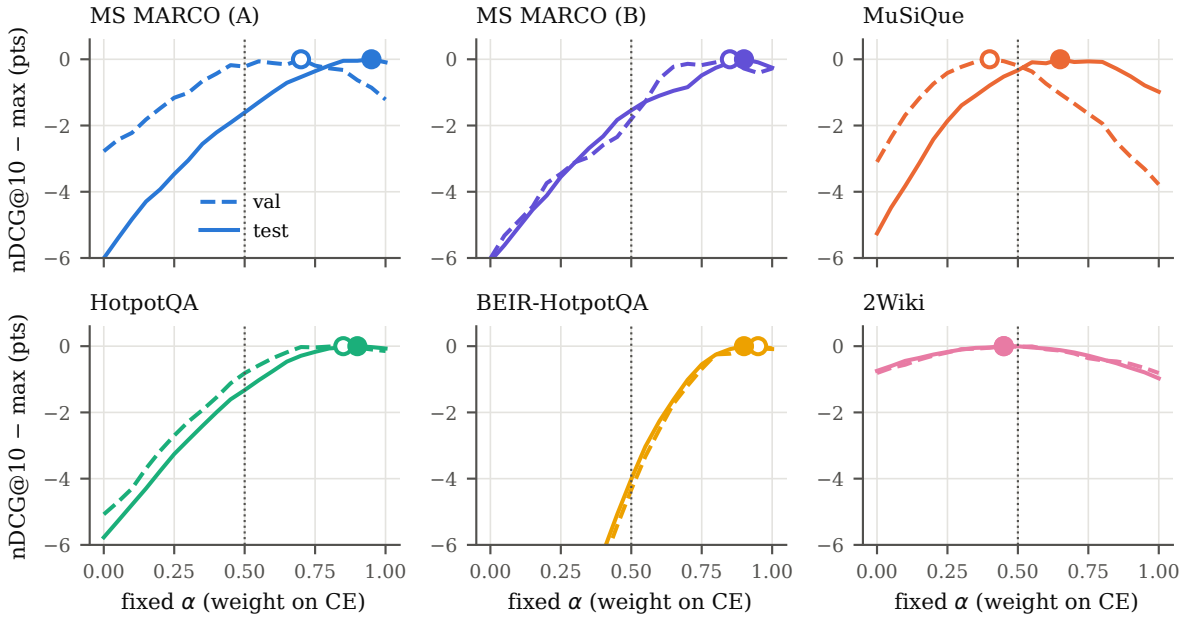


Figure 1: nDCG@10 as a function of the fixed weight α (min–max), relative to each curve's maximum; dashed = validation, solid = test, circles = arg-max. Left of the dotted line the contextual score has more weight. The single-hop reference (MS MARCO) peaks far to the right; MuSiQue (validation) and 2Wiki peak left of 0.5. The y-axis is clipped at -6 points.

	Test nDCG@10 (×100)			Best fixed α on val, by normaliser				min–max α^*		
Dataset	CE	GAT	CE−GAT	none	min–max	z-score	rank	val	test	favours
MS MARCO (A)	41.3	35.4	+5.9	0.05	0.70	0.70	0.55	**0.70**	0.95	sem/sem
MS MARCO (B)	41.5	35.7	+5.8	0.15	0.85	0.85	1.00	**0.85**	0.90	sem/sem
MuSiQue	71.1	66.8	+4.3	0.05	0.40	0.45	0.45	**0.40**	0.65	ctx/sem
HotpotQA	89.8	84.1	+5.7	0.10	0.85	0.85	0.90	**0.85**	0.90	sem/sem
BEIR-HotpotQA	72.6	55.5	+17.1	0.35	0.95	0.95	1.00	**0.95**	0.90	sem/sem
2Wiki	90.8	91.0	-0.2	0.05	0.45	0.50	0.55	**0.45**	0.45	ctx/ctx

Table 2: Dataset-level test of the hypothesis. α weights the cross-encoder: fused $= \alpha\, s_{sem} + (1-\alpha)\, s_{ctx}$. α^* is the arg-max of mean nDCG@10 over the grid $\{0, 0.05, \ldots, 1\}$; the val value (bold) is the one used by the fixed baseline, the test value is for analysis only. "favours": ctx if $\alpha^* < 0.5$, sem if $\alpha^* > 0.5$ (val/test). With raw scores (none) α mostly reflects the scale ratio of the two scores and is not interpretable.

random, title-link or hybrid graph leaves the pattern intact (Figure 3; Table 19 in Appendix B). Across the 14 normaliser/graph/split settings the 4-hop optimum is the lowest and below 0.5 in all 14, and the decline from 2 to 4 hops is monotone in 13.

- **Per question**, among questions whose ranking depends on α, the share whose optimal interval

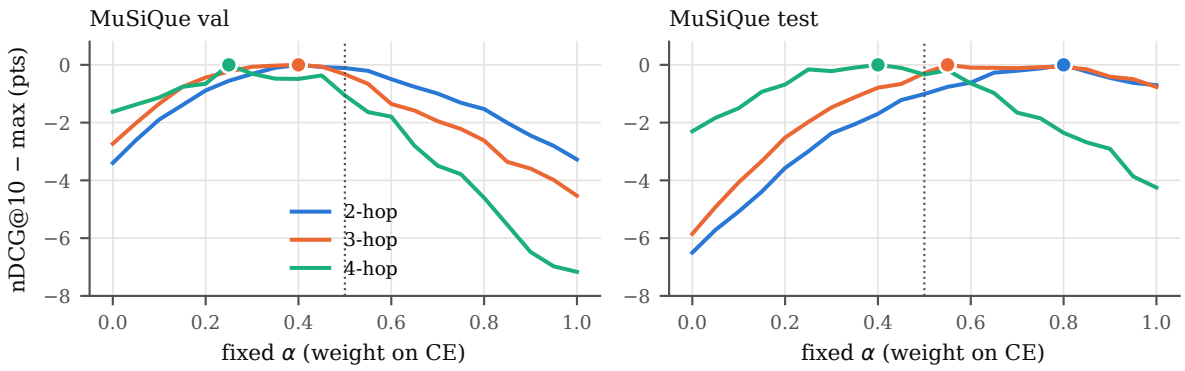


Figure 2: MuSiQue by hop count: nDCG@10 of the fixed fusion as a function of α (min–max) for 2-, 3- and 4-hop questions, relative to each curve's maximum; dots mark the optimum. The optimum moves left (towards the contextual score) as the number of hops grows, on validation (left) and test (right).

		nDCG@10 of each signal alone				Best fixed α						% prefer ctx	
		val		test		val			test				
Hops	n (test)	CE	GAT	CE	GAT	mm	z	rank	mm	z	rank	val	test
2-hop	1,252	76.9	76.8	73.8	68.0	**0.40**	0.50	**0.45**	0.80	0.80	0.95	48	40
3-hop	760	69.2	71.0	71.8	66.7	**0.40**	**0.40**	**0.30**	0.55	0.65	0.85	54	39
4-hop	405	66.1	71.7	61.3	63.3	**0.25**	**0.30**	**0.00**	**0.40**	**0.40**	**0.00**	60	52

Table 3: MuSiQue by hop count. Left: the cross-encoder loses 10.8 (val) and 12.5 (test) nDCG@10 points from 2 to 4 hops, the GAT only 5.1 and 4.7, and overtakes it on 4-hop questions. Middle: the best fixed α falls with the hop count under every scale-free normaliser (mm = min–max, z = z-score); bold = below 0.5, i.e. the contextual score dominates. Right: share of α-sensitive questions whose optimal α-interval is centred below 0.5. Tie-robust per-question test of the trend: Spearman(preferred α, hops) = -0.072 [-0.110, -0.032] on val ($p = 0.0005$) and -0.068 [-0.110, -0.025] on test ($p = 0.004$).

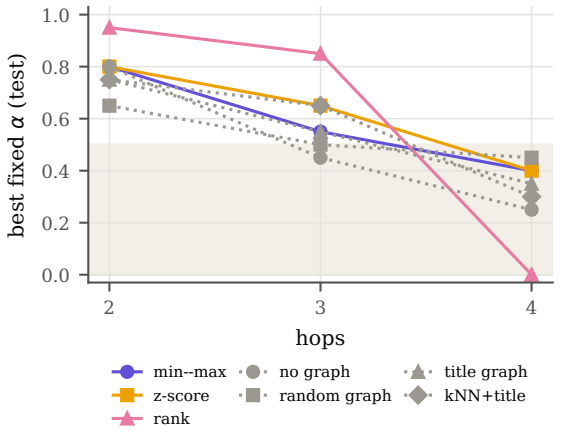


Figure 3: MuSiQue: best fixed α on test per hop count under the three scale-free normalisers (coloured) and for GATs retrained on four other graphs (grey, dotted; min–max). Shaded: $\alpha < 0.5$. Every setting declines from 2 to 4 hops, and every 4-hop optimum is below 0.5.

lies on the contextual side grows from 48% (2 hops) to 54% (3) and 60% (4) on validation, and from 40% to 52% on test. The Spearman correlation between a question's preferred α and its hop count is -0.072 [-0.110, -0.032] on validation ($p = 0.0005$) and -0.068 [-0.110, -0.025] on test ($p = 0.004$).

- **Finer composition types** agree: among MuSiQue's composition types with at least 100 questions, the 4-hop type 4hop1 has the lowest optimum on both splits (0.30 on validation and test).

4.3 Why: the cross-encoder degrades faster with hops

Figure 4 explains the trend. A cross-encoder scores each passage in isolation, so a passage that is relevant only through a bridge entity mentioned in another passage is hard for it; the more hops, the more such passages. Its nDCG@10 drops from 73.8 (2 hops) to 61.3 (4 hops) on test, whereas the graph-contextual score, whose passage representation is refined by attending to the other candidates, drops only from 68.0 to 63.3 and overtakes the cross-encoder. On validation the GAT

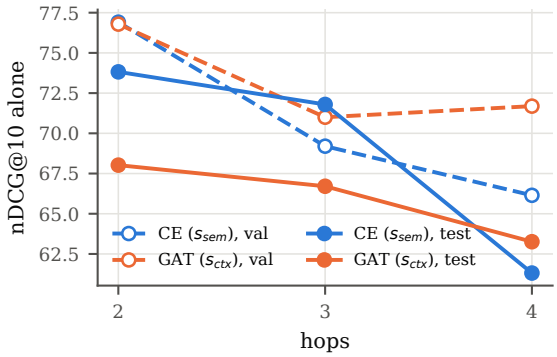


Figure 4: MuSiQue: nDCG@10 of each signal alone by hop count (dashed = validation, solid = test). The cross-encoder (CE) loses 10.8/12.5 points from 2 to 4 hops, the graph-contextual score (GAT) only 5.1/4.7, and the GAT overtakes the CE on 3- and 4-hop questions (validation) and on 4-hop questions (test).

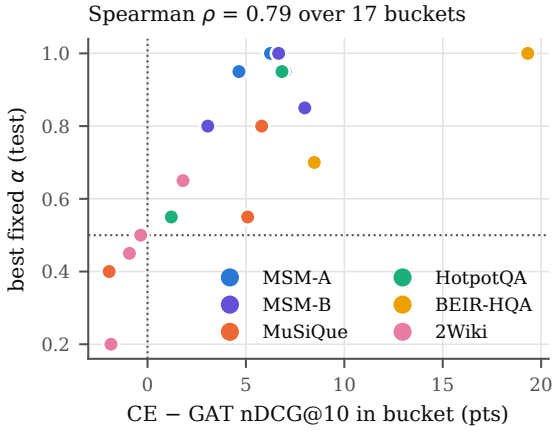


Figure 5: Each point is one question group (test): the CE−GAT nDCG@10 gap of its questions vs. its best fixed α (min–max). Groups in which the contextual score is at least as reliable as the cross-encoder (MuSiQue 4-hop and the three non-comparison 2Wiki types) all have $\alpha < 0.5$.

already matches the cross-encoder on 2-hop questions (76.8 vs. 76.9) and beats it by 1.8 and 5.6 points on 3- and 4-hop questions, which is why every hop count favours context there.

The same relation holds across all collections (Figure 5): over 17 question groups from the six test sets, the optimal α is strongly rank-correlated with the CE−GAT gap of the group's questions ($\rho = 0.79$). Complex questions therefore lower the optimal weight on the cross-encoder *because* they are the questions on which contextual evidence becomes the more reliable signal. Improving the contextual signal moves the weight in the same direction: on HotpotQA, adding title-

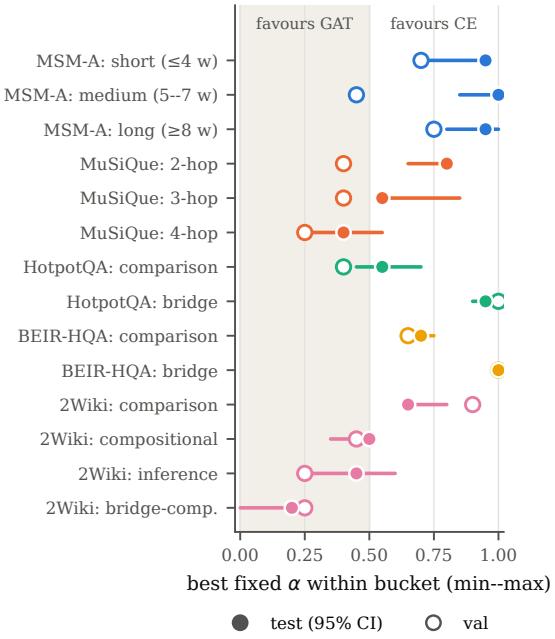


Figure 6: Best fixed α within each question group (min–max) for all collections. Filled: test with 95% bootstrap CI; open: validation. Shaded: the contextual score has more weight.

link edges between paragraphs raises GAT-only nDCG@10 from 84.1 to 87.4, lowers the optimal α from 0.85 to 0.65 and yields the best HotpotQA system (90.52; Table 20).

4.4 Question type on 2WikiMultiHopQA

2Wiki labels each question with one of four reasoning types (Table 4). Comparison questions name both entities, so each gold paragraph can be recognised on its own; the other types require following a relation from one paragraph to the next. As the hypothesis predicts, comparison questions favour the cross-encoder ($\alpha^* = 0.90$ on validation, 0.65 on test), while compositional (0.45/0.50), inference (0.25/0.45) and bridge-comparison questions (0.25/0.20) favour the contextual score; for bridge-comparison questions, which combine a bridge step with a comparison, 67% of the α-sensitive questions prefer the contextual side. The contextual score outperforms the cross-encoder on exactly the three relational types.

Importantly, **a learned gate recovers this ordering without any label**. The query-only Q gate, which sees nothing but the query embedding, assigns comparison questions the highest mean α under all three normalisers (0.72 / 0.52 / 0.59 for min–max / z-score / rank) and the relational

	nDCG@10 (test)		Best fixed α		% prefer	Learned Q gate: mean α and ΔnDCG@10 vs. fixed (test)					
Type	CE	GAT	val	test	ctx	min–max		z-score		rank	
comparison	97.3	95.5	0.90	0.65	30	0.72	+0.10	0.52	-0.04	0.59	-0.07
compositional	88.5	88.9	**0.45**	0.50	53	0.45	-0.01	0.34	-0.08	0.44	+0.25$^{\blacktriangle}$
inference	86.7	87.6	**0.25**	**0.45**	59	0.30	-0.23$^{\triangledown}$	0.22	-0.03	0.38	+0.27
bridge-comparison	90.1	92.0	**0.25**	**0.20**	67	0.35	+0.16$^{\blacktriangle}$	0.33	+0.22$^{\blacktriangle}$	0.17	+0.81$^{\blacktriangle}$

Table 4: 2WikiMultiHopQA by question type, ordered from the simplest (comparison: both entities named in the question) to the most compositional (bridge-comparison). Bold $\alpha < 0.5$. The query-only gate, which never sees the type label, gives comparison questions the highest α under all three normalisers, and gains significantly on bridge-comparison questions ($^{\blacktriangle}$/$^{\triangledown}$: paired bootstrap $p < 0.05$).

types lower values (down to 0.17–0.35 for bridge-comparison). It translates into significant gains exactly where the hypothesis says context matters most: on bridge-comparison questions the Q gate improves over the fixed weight under every normaliser (+0.16, +0.22 and +0.81 nDCG@10). Figure 6 summarises the group optima across all collections.

4.5 Query-dependent α versus a fixed α

A type-aware weight beats a single weight. The direct way to make α query-dependent is to give each question type its own weight. Choosing one α per type on validation and applying it on test (Table 7) significantly improves on the single global weight on all three collections with type labels: HotpotQA +0.14 [+0.02, +0.25], BEIR-HotpotQA +0.42 [+0.30, +0.53] and 2Wiki +0.07 [+0.01, +0.12] nDCG@10. The selected weights follow the hypothesis on 2Wiki (comparison 0.90; bridge-comparison and inference 0.25).

Learned gates gain mainly on multi-hop data. Table 6 compares every learned gate with the best fixed α under the same normaliser. Of the 11 significant improvements, 7 are on multi-hop collections, and they are also the largest: +1.37, +1.22 and +0.43 on MuSiQue and +0.41, +0.30 and +0.19 on 2Wiki (rank normalisation), and +0.18 on BEIR-HotpotQA (z-score Q, the best BEIR-HotpotQA system overall at 72.87). The four MS MARCO improvements are all ≤ 0.05 points. On MuSiQue the rank-normalised gates improve every hop count (S: +1.64, +1.10 and +1.07 for 2, 3 and 4 hops; all $p < 0.05$), and on 2Wiki they gain most on bridge-comparison questions (Table 4).

A single gate trained on mixed data separates simple from complex queries. When one Q gate is trained on MS MARCO, MuSiQue and HotpotQA together, without any dataset indicator (§4.6), it assigns the lowest weight to the cross-encoder on MuSiQue (mean α 0.54), more on MS MARCO (0.59) and the most on HotpotQA (0.68), with substantial per-query variation (sd 0.12–0.16; Figure 7). On MuSiQue this gate scores 71.77 nDCG@10, above the in-domain fixed weight (71.28), although a fixed $\alpha = 0.55$ trained on the same mixed data scores 71.98.

More headroom on multi-hop questions. The per-query oracle shows how much a perfect query-dependent weight could add (Table 5): 7.4 nDCG@10 points on MuSiQue against 5.0 on MS MARCO. The difference, +2.4 [+1.9, +2.9], is significant after Holm correction (pre-registered H3; Appendix C), and the headroom is largest for 4-hop questions (7.9 points).

4.6 Transfer and mixed training

Table 9 lists the transfer experiments. The mixed-data gate is discussed above. A gate trained only on MS MARCO and applied zero-shot keeps MS MARCO's weight ($\alpha \approx 0.70$) and does not adapt; it therefore cannot recover the lower weight that multi-hop questions need. Training on MS MARCO and HotpotQA and testing on MuSiQue (leave-one-out) likewise keeps $\alpha \approx 0.70$, since neither training source contains questions with three or four hops.

On 2WikiMultiHopQA the zero-shot gate (90.99) stays clearly below the in-domain fixed weight (91.75). That in-domain weight is context-leaning ($\alpha_0 = 0.45$) and is chosen consistently on validation and test. 2Wiki pools are closed and small, so even BM25 reaches 87.77 and the cross-encoder alone 90.77. Against that baseline the in-domain fusion adds 0.98 points, while the transferred gate, with $\alpha = 0.70$ and the MS MARCO GAT, adds only 0.22.

	MSM-A		MSM-B		MuSiQue		HotpotQA		BEIR-HQA		2Wiki	
	nDCG	MRR	nDCG	MRR	nDCG	MRR	nDCG	MRR	nDCG	MRR	nDCG	MRR
BM25 (first stage / in-pool)	22.84	18.40	22.96	18.51	55.11	75.70	81.74	87.87	63.29	80.32	87.77	96.51
SBERT (s_{bi})	35.30	30.38	35.74	30.86	67.08	81.83	82.94	87.19	53.67	70.29	89.19	96.89
CE only (α=1)	41.34	36.39	41.54	36.58	71.09	87.32	89.84	95.42	72.61	89.02	90.77	98.16
GAT only (α=0)	35.42	30.43	35.70	30.58	66.81	81.48	84.14	88.39	55.47	72.63	91.00	97.54
ReGAT-BERT-style (a, b; BCE)	39.83$^{\triangledown}$	34.72$^{\triangledown}$	40.25$^{\triangledown}$	35.22$^{\triangledown}$	71.74$^{\blacktriangle}$	87.29$^{\blacktriangle}$	89.89	95.35	68.81$^{\triangledown}$	85.35$^{\triangledown}$	91.73	98.56
Fixed α (grid on val)	40.89	35.86	41.69	36.84	71.28	86.28	89.87	95.28	72.63	89.03	91.75	98.54
Fixed α (learned)	40.89	35.86	41.69	36.84	71.28	86.28	89.89	95.32	72.63	89.03	91.75	98.54
Adaptive $\alpha(q)$ Q	40.89$^{\blacktriangle}$	35.87	41.70	36.83	71.28	86.28	89.87	95.28	72.63	89.03	91.77	98.37$^{\triangledown}$
Adaptive $\alpha(q, s)$ QS	40.89	35.86	41.70	36.84	71.27	86.28	89.91	95.49	72.63	89.03	91.63$^{\triangledown}$	98.40$^{\triangledown}$
Adaptive $\alpha(s)$ S	40.90	35.87	41.71$^{\blacktriangle}$	36.86	71.27	86.31	89.93	95.49$^{\blacktriangle}$	72.64	89.03	91.75	98.53
Adaptive QS+C	–	–	–	–	71.28	86.28	89.92	95.52$^{\blacktriangle}$	72.63	89.03	91.63$^{\triangledown}$	98.35$^{\triangledown}$
Oracle per-query α (bound)	45.90	41.81	46.01	41.95	78.72	92.50	92.86	97.69	75.56	92.04	94.00	99.49

Table 5: Main results on the test sets (nDCG@10 and MRR@10, $\times 100$, mean over 5 seeds) under the primary normaliser (min–max, fixed on MS MARCO val). $^{\blacktriangle}/^{\triangledown}$: significantly better/worse than the best fixed α (paired bootstrap over queries, 10k resamples, $p < 0.05$, uncorrected). The oracle row picks α per query on the test labels and is an upper bound, not a method. QS+C was added with the multi-hop extension and not run on MS MARCO (–). Full metrics at $k \in \{1, 3, 5, 10, 50\}$ are in Appendix A.

Norm.	Gate	MSM-A	MSM-B	MuSiQue	HotpotQA	BEIR-HQA	2Wiki
none	Q	-0.48$^{\triangledown}$	+0.01	+0.00	-0.00	-0.00	+0.00
	QS	-0.48$^{\triangledown}$	+0.00	+0.00	+0.01	-0.00$^{\triangledown}$	+0.00
	S	-0.49$^{\triangledown}$	+0.00	+0.02	+0.01	-0.00	-0.00
minmax	Q	**+0.00**$^{\blacktriangle}$	+0.00	+0.00	-0.02	-0.00	+0.03
	QS	+0.00	+0.01	-0.01	+0.02	+0.00	-0.12$^{\triangledown}$
	S	+0.00	**+0.02**$^{\blacktriangle}$	-0.00	+0.04	+0.01	+0.00
	QS+C	–	–	+0.00	+0.03	-0.00	-0.11$^{\triangledown}$
zscore	Q	+0.01	**+0.05**$^{\blacktriangle}$	-0.15$^{\triangledown}$	+0.06	**+0.18**$^{\blacktriangle}$	+0.00
	QS	**+0.02**$^{\blacktriangle}$	-0.01	-0.14$^{\triangledown}$	+0.00	-0.01$^{\triangledown}$	-0.00
	S	+0.01	+0.01	-0.12$^{\triangledown}$	+0.01	-0.01	-0.01$^{\triangledown}$
rank	Q	+0.01	+0.02	**+0.43**$^{\blacktriangle}$	-0.01	-0.01	**+0.30**$^{\blacktriangle}$
	QS	-0.12	-0.04	**+1.22**$^{\blacktriangle}$	-0.04	-0.04$^{\triangledown}$	**+0.41**$^{\blacktriangle}$
	S	-0.10	-0.02	**+1.37**$^{\blacktriangle}$	-0.00	-0.02$^{\triangledown}$	**+0.19**$^{\blacktriangle}$

Table 6: Every adaptive gate against the best fixed α under the same normaliser: test ΔnDCG@10 $\times 100$ (paired bootstrap; $^{\blacktriangle}$ significantly better, $^{\triangledown}$ worse, $p < 0.05$ uncorrected).

4.7 Boundary conditions

HotpotQA. On HotpotQA and BEIR-HotpotQA the cross-encoder keeps the weight ($\alpha^* = 0.85$–0.95), and bridge questions favour it even more than comparison questions (Figure 6). HotpotQA questions have exactly two gold paragraphs, and many can be answered by matching the question against a single paragraph (Min et al., 2019); consistently, the cross-encoder leads the GAT by 6.8 (distractor) and 19.3 (BEIR) points on bridge questions. The hypothesis therefore applies to questions whose hops are genuinely necessary, as in MuSiQue and 2Wiki, rather than to every question labelled multi-hop. Better graphs narrow the gap here too: title links lower the optimal α on HotpotQA from 0.85 to 0.65 (Table 20).

End-to-end gates under min–max. Under the primary normaliser the learned gates gain at most 0.04 points over the fixed weight (Table 5). Table 8 shows why: on MS MARCO, MuSiQue and BEIR-HotpotQA early stopping returns the initialisation in every run, so the gate equals the fixed weight; on MuSiQue that fixed weight ($\alpha_0 = 0.40$) is itself already on the contextual side. The benefit of query dependence is realised by type-aware weights, rank-normalised gates and mixed training rather than by the min–max gates. The preregistered H1/H2 tests on learned gates are correspondingly inconclusive or rest on near-constant gates (Appendix C).

Validation versus test. MuSiQue and MS MARCO A validation sets are drawn from training questions; the overall MuSiQue test optimum

Data	Buckets	α_0	Per-bucket α (chosen on val)	Δ	95% CI
MSM-A	length	0.70	long ($\geq$8 w) 0.75; medium (5–7 w) 0.45; short ($\leq$4 w) 0.70	-0.69$^{\triangledown}$	[-0.87, -0.52]
MSM-A	wh-word	0.70	how 0.85; other 0.55; what 0.45; when 0.85; where 0.85; who 0.85; why 0.85	-0.79$^{\triangledown}$	[-1.01, -0.57]
MSM-B	length	0.85	long ($\geq$8 w) 1.00; medium (5–7 w) 0.85; short ($\leq$4 w) 0.75	-0.03	[-0.30, +0.24]
MuSiQue	length	0.40	long ($\geq$8 w) 0.40; medium (5–7 w) 0.15	-0.01	[-0.02, +0.00]
MuSiQue	hops (gold)	0.40	2-hop 0.40; 3-hop 0.40; 4-hop 0.25	-0.03	[-0.14, +0.08]
HotpotQA	length	0.85	long ($\geq$8 w) 0.80; medium (5–7 w) 0.85	-0.03	[-0.09, +0.02]
HotpotQA	type (gold)	0.85	bridge 1.00; comparison 0.40	+0.14$^{\blacktriangle}$	[+0.02, +0.25]
BEIR-HQA	length	0.95	long ($\geq$8 w) 0.95; medium (5–7 w) 1.00	-0.00	[-0.01, +0.00]
BEIR-HQA	type (gold)	0.95	bridge 1.00; comparison 0.65	+0.42$^{\blacktriangle}$	[+0.30, +0.53]
2Wiki	length	0.45	long ($\geq$8 w) 0.45; medium (5–7 w) 0.20	+0.00	[-0.02, +0.02]
2Wiki	type (gold)	0.45	bridge-comp. 0.25; comparison 0.90; compositional 0.45; inference 0.25	+0.07$^{\blacktriangle}$	[+0.01, +0.12]

Table 7: Bucketed fixed α: one α per bucket selected on val and applied on test, against the single global α_0 (test ΔnDCG@10 $\times$100, paired bootstrap). Length buckets are available at test time; gold hop/type labels are not, so those rows measure how much a type-aware α *could* give (analysis only).

		In-domain gate: mean (sd over test queries) of learned α				Restored	Mixed gate
Dataset	α_0	Q	QS	S	QS+C	at init	Q
MS MARCO (A)	0.70	0.700 (0.000)	0.700 (0.000)	0.699 (0.002)	–	100%	0.59 (0.12)
MS MARCO (B)	0.85	0.857 (0.002)	0.853 (0.001)	0.850 (0.001)	–	80%	–
MuSiQue	0.40	0.400 (0.000)	0.400 (0.001)	0.400 (0.002)	0.400 (0.001)	100%	0.54 (0.16)
HotpotQA	0.85	0.850 (0.000)	0.883 (0.210)	0.919 (0.081)	0.909 (0.138)	30%	0.68 (0.15)
BEIR-HotpotQA	0.95	0.950 (0.000)	0.950 (0.000)	0.950 (0.000)	0.950 (0.000)	100%	–
2Wiki	0.45	0.474 (0.237)	0.511 (0.126)	0.449 (0.002)	0.493 (0.157)	55%	–

Table 8: What the gates learn (min–max, $\lambda = 0.01$). The gates are initialised at α_0 and early-stopped on val nDCG@10; "restored at init" is the share of adaptive runs (all gates, 5 seeds) whose best val epoch was the initialisation, i.e. the returned model is the fixed baseline. Mixed gate: one Q gate trained on MS MARCO + MuSiQue + HotpotQA (§4.6), its α on each test set.

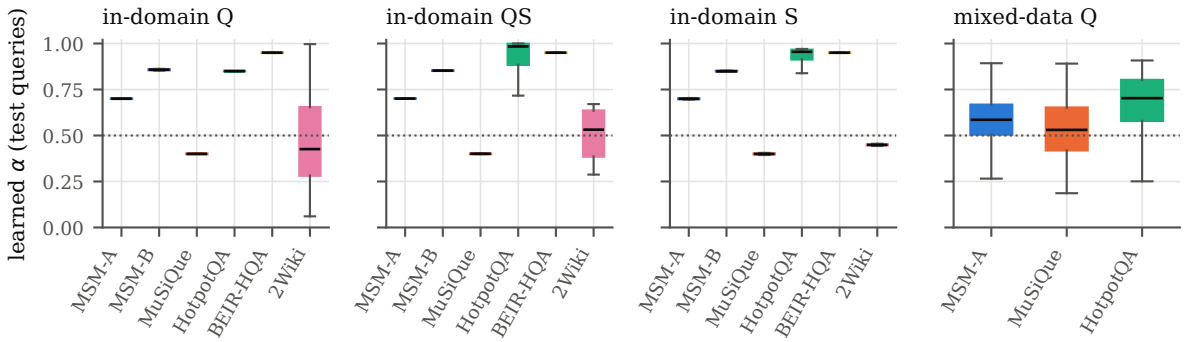


Figure 7: Distribution of the learned α over test questions (seed-averaged; min–max). The 2Wiki gates spread α over a wide range, and the gate trained on mixed data places MuSiQue lowest. Under min–max several in-domain gates stay at their initialisation α_0, which is already below 0.5 on MuSiQue and 2Wiki.

(0.65) is therefore above its validation optimum (0.40). The hop-count trend, the main evidence for the hypothesis, holds on both splits.

			Transferred gate, test nDCG@10				In-domain		Q gate
Setting	Train	Test	fixed	Q	QS	S	fixed	α_0	$\bar{\alpha}$ (sd)
leave-one-out	MSM+HQA	MuSiQue	71.99	71.93	71.98	72.00	71.28	0.70	0.73 (0.02)
mixed	MSM+MuSiQue+HQA	MS MARCO (A)	40.13	40.29	40.18	40.22	40.89	0.55	0.59 (0.12)
mixed	MSM+MuSiQue+HQA	HotpotQA	88.88	89.44	89.32	89.34	89.89	0.55	0.68 (0.15)
mixed	MSM+MuSiQue+HQA	MuSiQue	71.98	71.77	71.81	71.79	71.28	0.55	0.54 (0.16)
zero-shot	MSM	BEIR-HotpotQA	71.45	71.45	71.45	71.46	72.63	0.70	0.70 (0.00)
zero-shot	MSM	HotpotQA	89.24	89.24	89.25	89.20	89.89	0.70	0.70 (0.00)
zero-shot	MSM	2Wiki	90.99	90.99	90.99	90.99	91.75	0.70	0.70 (0.00)

Table 9: Transfer and mixed training of the gate (E4, min–max, $\times 100$, 5 seeds). Zero-shot uses the frozen MS MARCO GAT for s_{ctx}; mixed and leave-one-out use each dataset's own GAT. α_0 is the fixed α the gate starts from (selected on the training sources' val).

Claim	Supporting evidence	Status
$\alpha > 0.5$ for single-hop queries	MS MARCO 0.70–0.95 under all scale-free normalisers	supported
$\alpha < 0.5$ for multi-hop queries	MuSiQue 0.40 (val), all hop counts < 0.5 on val; 4-hop < 0.5 in 14/14 settings; 2Wiki 0.45; 2Wiki relational types 0.20–0.50	supported for MuSiQue, 2Wiki
α falls with complexity	MuSiQue 0.80/0.55/0.40 (test), 0.40/0.40/0.25 (val), monotone in 13/14 settings; per-question $\rho < 0$, $p \leq 0.004$; 2Wiki comparison $>$ relational types	supported
Mechanism	CE loses 12.5 points from 2 to 4 hops, GAT 4.7; $\rho = 0.79$ between CE$-$GAT gap and α^*	supported
Query-dependent $\alpha >$ fixed	type-aware α: +0.14, +0.42, +0.07 (sig.); 7 of 11 significant gate gains on multi-hop (up to +1.37); 2Wiki Q gate recovers the type ordering; oracle headroom 7.4 (MuSiQue) vs. 5.0	supported where structure is exploited
Not observed	HotpotQA / BEIR-HotpotQA ($\alpha^* \geq 0.85$); min–max gates stay near α_0	boundary

Table 10: Summary of the evidence for the hypothesis.

5 Discussion

Table 10 summarises the evidence. The central prediction, that complex multi-hop questions should shift the fusion weight from the cross-encoder to the graph-contextual score, is borne out on the collection designed to require every hop. On MuSiQue the optimum falls step by step from 2 to 4 hops in 13 of 14 normaliser/graph/split settings, and reaches the contextual side for 4-hop questions on both splits. On 2Wiki the reasoning type orders the optimal weights as predicted, and a gate that sees only the query learns that ordering. The mechanism is intuitive and measurable: a cross-encoder judges each passage alone and loses accuracy quickly as hops accumulate, while a score refined over the candidate graph degrades gracefully, so the relative reliability of the two signals, and with it the optimal weight, shifts towards context.

Making α query-dependent is worthwhile to the extent that the dependence is captured. Coarse type-aware weights deliver consistent significant gains, learned gates deliver their largest gains on multi-hop data, and a single gate trained on mixed data already separates MuSiQue from MS MARCO and HotpotQA. The 7.4-point oracle headroom on MuSiQue shows that much more is available. Two directions follow directly: gates that predict question complexity (e.g. the number of hops) explicitly, trained with objectives that match the evaluation metric, and validation data drawn from the target distribution so that early stopping can reward adaptation.

6 Conclusion

We asked whether the weight between a cross-encoder and a graph-contextual score should depend on query complexity. It should: the optimal weight is above 0.5 on single-hop MS MARCO, below 0.5 on MuSiQue and 2WikiMultiHopQA, and it decreases with each additional hop on MuSiQue, because the cross-encoder loses reliability faster than the contextual score as questions require more hops. Query-dependent weighting improves retrieval where it captures this structure, through type-aware weights and learned gates on multi-hop data, and the large oracle headroom on multi-hop questions leaves ample room for gates

that model complexity directly.

Limitations

All models are small MiniLM checkpoints trained on MS MARCO and never fine-tuned on multi-hop data; a cross-encoder trained on multi-hop data would likely raise every optimum. Three multi-hop settings are closed pools in which the gold paragraphs are always present, so they test reranking, not retrieval. The graphs are per-query candidate graphs; ReGAT-BERT's original structural edges cannot be built on MS MARCO, and our results concern this adaptation. On HotpotQA and BEIR-HotpotQA the effect is not observed (§4.7). Under the primary min–max normaliser the end-to-end gates add at most 0.04 nDCG@10 points, and early stopping often returns their initialisation; the larger gate gains occur under rank normalisation, whose fixed baseline is weaker. The validation sets of MS MARCO A and the closed-pool datasets are carved from training questions, which shifts the overall MuSiQue optimum between validation and test. Group optima are computed on test data for analysis only and are reported together with their validation counterparts; type-aware weights use gold labels that would have to be predicted at test time. The per-question correlation with hop count is small in magnitude ($|\rho| \approx 0.07$), as most questions are indifferent to α over a wide range. Significance marks in the ablation grid are uncorrected; the nine pre-registered tests are Holm-corrected (Appendix C).

Ethics Statement

We use public research datasets under their licences and release no new data. The work does not involve human subjects or personal data beyond what these benchmarks already contain.

References

n.d. ReGAT-BERT: graph-attention contextual scoring fused with a BERT cross-encoder for passage ranking. Full bibliographic reference to be completed.

Akari Asai, Kazuma Hashimoto, Hannaneh Hajishirzi, Richard Socher, and Caiming Xiong. 2020. Learning to retrieve reasoning paths over Wikipedia graph for question answering. In *International Conference on Learning Representations*.

Shaked Brody, Uri Alon, and Eran Yahav. 2022. How attentive are graph attention networks? In *International Conference on Learning Representations*.

Zhe Cao, Tao Qin, Tie-Yan Liu, Ming-Feng Tsai, and Hang Li. 2007. Learning to rank: From pairwise approach to listwise approach. In *Proceedings of the 24th International Conference on Machine Learning*, pages 129–136.

Gordon V. Cormack, Charles L. A. Clarke, and Stefan Büttcher. 2009. Reciprocal rank fusion outperforms Condorcet and individual rank learning methods. In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 758–759.

Nick Craswell, Bhaskar Mitra, Emine Yilmaz, and Daniel Campos. 2021. Overview of the TREC 2020 deep learning track. *arXiv preprint arXiv:2102.07662*.

Nick Craswell, Bhaskar Mitra, Emine Yilmaz, Daniel Campos, and Ellen M. Voorhees. 2020. Overview of the TREC 2019 deep learning track. *arXiv preprint arXiv:2003.07820*.

Fernando Diaz. 2005. Regularizing ad hoc retrieval scores. In *Proceedings of the 14th ACM International Conference on Information and Knowledge Management*, pages 672–679.

Bradley Efron and Robert J. Tibshirani. 1993. *An Introduction to the Bootstrap*. Chapman & Hall.

Matthias Fey and Jan Eric Lenssen. 2019. Fast graph representation learning with PyTorch Geometric. In *ICLR Workshop on Representation Learning on Graphs and Manifolds*.

Edward A. Fox and Joseph A. Shaw. 1994. Combination of multiple searches. In *Proceedings of the Second Text REtrieval Conference (TREC-2)*, pages 243–252.

Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. 2020. Constructing a multi-hop QA dataset for comprehensive evaluation of reasoning steps. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 6609–6625.

Oren Kurland and Lillian Lee. 2005. PageRank without hyperlinks: Structural re-ranking using links induced by language models. In *Proceedings of the 28th Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 306–313.

Jimmy Lin, Xueguang Ma, Sheng-Chieh Lin, Jheng-Hong Yang, Ronak Pradeep, and Rodrigo Nogueira. 2021. Pyserini: A Python toolkit for reproducible information retrieval research with sparse and dense representations. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 2356–2362.

Ilya Loshchilov and Frank Hutter. 2019. Decoupled weight decay regularization. In *International Conference on Learning Representations*.

Sean MacAvaney, Craig Macdonald, and Iadh Ounis. 2022a. Streamlining evaluation with ir-measures. In *Advances in Information Retrieval: 44th European Conference on IR Research*, pages 305–310.

Sean MacAvaney, Nicola Tonellotto, and Craig Macdonald. 2022b. Adaptive re-ranking with a corpus graph. In *Proceedings of the 31st ACM International Conference on Information and Knowledge Management*, pages 1491–1500.

Sewon Min, Eric Wallace, Sameer Singh, Matt Gardner, Hannaneh Hajishirzi, and Luke Zettlemoyer. 2019. Compositional questions do not necessitate multi-hop reasoning. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 4249–4257.

Mark Montague and Javed A. Aslam. 2001. Relevance score normalization for metasearch. In *Proceedings of the Tenth International Conference on Information and Knowledge Management*, pages 427–433.

Tri Nguyen, Mir Rosenberg, Xia Song, Jianfeng Gao, Saurabh Tiwary, Rangan Majumder, and Li Deng. 2016. MS MARCO: A human generated MAchine reading COmprehension dataset. In *Proceedings of the Workshop on Cognitive Computation (CoCo@NIPS)*.

Rodrigo Nogueira and Kyunghyun Cho. 2019. Passage re-ranking with BERT. *arXiv preprint arXiv:1901.04085*.

Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, pages 3982–3992.

Stephen Robertson and Hugo Zaragoza. 2009. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4):333–389.

Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. 2021. BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models. In *Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.

Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2022. MuSiQue: Multihop questions via single-hop question composition. *Transactions of the Association for Computational Linguistics*, 10:539–554.

Aaron van den Oord, Yazhe Li, and Oriol Vinyals. 2018. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*.

Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. 2018. Graph attention networks. In *International Conference on Learning Representations*.

Wenhui Wang, Furu Wei, Li Dong, Hangbo Bao, Nan Yang, and Ming Zhou. 2020. MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers. In *Advances in Neural Information Processing Systems*, volume 33.

Wenhan Xiong, Xiang Lorraine Li, Srini Iyer, Jingfei Du, Patrick Lewis, William Yang Wang, Yashar Mehdad, Wen-tau Yih, Sebastian Riedel, Douwe Kiela, and Barlas Oğuz. 2021. Answering complex open-domain questions with multi-hop dense retrieval. In *International Conference on Learning Representations*.

Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. 2018. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2369–2380.

A Full results

Tables 11–16 report nDCG, MRR, Precision and Recall at $k \in \{1, 3, 5, 10, 50\}$ and MAP on every test set for the reference rows, all gates (with $\lambda = 0.01$ and $\lambda = 0$), the BCE loss ablation and the oracle. Precision@k divides by k; on closed pools the pool size bounds Recall@50 and Precision@50. Table 17 gives the out-of-distribution TREC DL 2019/2020 results (Craswell et al., 2020, 2021) for MS MARCO. Table 18 lists every question group with its optimum on both splits.

B Graph variants

Table 19 repeats the hop analysis for a GAT retrained on each graph type: the 4-hop optimum is the lowest and below 0.5 for every graph. Table 20 gives the effectiveness of each graph type: on HotpotQA, title links connect 68% of the gold paragraph pairs (vs. 48% for k-NN), raise GAT-only nDCG@10 by 3.3 points, lower α_0 to 0.65 and give the best HotpotQA system; the only significant gain of a learned gate in this table is on the hybrid graph (+0.18), which links 80% of the gold pairs. On MuSiQue the hop trend does not depend on the edges, indicating that the contextual signal there comes largely from the GAT's in-domain, listwise-trained passage representation. Table 21 gives the validation sensitivity of s_{ctx} to the graph parameters.

C Pre-registered hypothesis tests

Before any multi-hop run we fixed four hypotheses about the *learned* gates: H1, the gate's α decreases with complexity; H2, the adaptive gain exceeds MS MARCO's; H3, the oracle headroom exceeds MS MARCO's; H4, effects from the graph vanish with random or empty graphs. Table 22 reports the nine primary tests exactly as registered. H3 is supported on MuSiQue (+2.4 points more headroom than MS MARCO, Holm $p = 0.001$) and H1 on HotpotQA, but the HotpotQA gate moves in the opposite direction to its own nDCG optimum (it gives comparison questions a higher α, 0.90 vs. 0.88, whereas their optimum is lower). The H1 verdicts on MuSiQue and BEIR-HotpotQA concern gates whose α varies by less than 10^{-3} across queries and are not interpretable; H2 is inconclusive on all three datasets; H3 is contradicted on HotpotQA and BEIR-HotpotQA, which have less headroom than MS MARCO (3.0 and 2.9 points). Because these tests depend on whether a gate adapted at all, the main text relies on the gate-independent group optima.

D Step 0: MS MARCO diagnosis

On MS MARCO, the single-hop reference, the learned weight is stable at $\alpha \approx 0.70$ (sd 0.0003–0.006 across queries) and neither learned nor oracle α varies systematically with query length, wh-word or entity count (Spearman with length and entity count +0.02 to +0.08; Figure 8). This is what the hypothesis expects for a collection of uniformly simple questions: a single weight on the cross-encoder side suffices.

E Reproducibility

All numbers are produced by the released pipeline (`python -m src.pipeline`); this report is generated from its result files by `python -m src.report_analysis`, `python -m src.report_hops` and `python -m src.report_paper`. Software: PyTorch 2.15, PyTorch Geometric 2.8.1, sentence-transformers 6.2, Pyserini 1.6 (Lucene BM25 indexes `msmarco-v1-passage` and `beir-v1.0.0-hotpotqa.flat`), ir_datasets 0.6.3, ir_measures 0.4.3, spaCy 3.8 with `en_core_web_sm`. Hardware: one NVIDIA RTX 6000 Ada (48 GB). The multi-hop extension ran in 1 h 22 min, the MS MARCO study in under an hour. A golden regression confirmed that the variable-length (masked) code reproduces all 2×182 MS MARCO runs exactly (maximum per-query difference 3×10^{-8}). Deviations from the original protocol are logged in the repository's `NOTES.md`.

	nDCG@k					MRR@k					P@k					R@k					
	1	3	5	10	50	1	3	5	10	50	1	3	5	10	50	1	3	5	10	50	MAP
BM25 (first stage / in-pool)	10.5	17.0	19.7	22.8	27.4	10.5	15.6	17.1	18.4	19.4	10.5	7.5	5.9	3.9	1.2	10.2	21.7	28.5	37.9	58.1	19.2
SBERT (s_{bi})	20.4	29.3	32.4	35.3	38.2	20.4	27.5	29.2	30.4	31.1	20.4	12.3	9.0	5.4	1.4	19.8	35.7	43.2	52.0	64.4	30.6
CE only (α=1)	25.5	35.6	38.8	41.3	43.1	25.5	33.6	35.4	36.4	36.8	25.5	14.8	10.5	6.1	1.4	24.7	42.9	50.5	58.2	65.4	36.2
GAT only (α=0)	20.4	29.2	32.4	35.4	38.2	20.4	27.4	29.2	30.4	31.1	20.4	12.3	9.0	5.5	1.4	19.8	35.6	43.3	52.4	64.5	30.6
ReGAT-BERT-style (a, b; BCE)	23.9	33.8	37.1	39.8	41.8	23.9	31.8	33.6	34.7	35.2	23.9	14.2	10.2	6.0	1.4	23.2	41.0	48.8	57.1	65.4	34.6
Fixed α (grid on val)	25.0	35.1	38.4	40.9	42.7	25.0	33.0	34.8	35.9	36.3	25.0	14.7	10.5	6.0	1.4	24.3	42.4	50.3	57.9	65.4	35.7
Fixed α (learned)	25.0	35.1	38.4	40.9	42.7	25.0	33.0	34.8	35.9	36.3	25.0	14.7	10.5	6.0	1.4	24.3	42.4	50.3	57.9	65.4	35.7
Adaptive $\alpha(q)$ Q	25.0	35.1	38.4	40.9	42.7	25.0	33.0	34.8	35.9	36.3	25.0	14.7	10.5	6.0	1.4	24.3	42.4	50.3	57.9	65.4	35.7
Adaptive $\alpha(q, s)$ QS	24.9	35.1	38.4	40.9	42.7	24.9	33.0	34.8	35.9	36.3	24.9	14.7	10.5	6.0	1.4	24.3	42.4	50.3	57.9	65.4	35.7
Adaptive $\alpha(s)$ S	25.0	35.1	38.4	40.9	42.7	25.0	33.1	34.8	35.9	36.3	25.0	14.7	10.5	6.0	1.4	24.3	42.4	50.2	57.9	65.4	35.7
Q, $\lambda = 0$	25.0	35.1	38.4	40.9	42.7	25.0	33.0	34.8	35.9	36.3	25.0	14.7	10.5	6.0	1.4	24.3	42.4	50.3	57.9	65.4	35.7
QS, $\lambda = 0$	24.9	35.1	38.4	40.9	42.7	24.9	33.0	34.8	35.9	36.3	24.9	14.7	10.5	6.0	1.4	24.3	42.4	50.3	57.9	65.4	35.7
S, $\lambda = 0$	25.0	35.1	38.4	40.9	42.7	25.0	33.1	34.8	35.9	36.3	25.0	14.7	10.5	6.0	1.4	24.3	42.4	50.2	57.9	65.4	35.7
Fixed α, BCE loss (minmax)	25.0	35.1	38.4	40.9	42.7	25.0	33.0	34.8	35.9	36.3	25.0	14.7	10.5	6.0	1.4	24.3	42.4	50.3	57.9	65.4	35.7
QS, BCE loss (minmax)	24.9	35.1	38.4	40.9	42.7	24.9	33.0	34.8	35.9	36.3	24.9	14.7	10.5	6.0	1.4	24.3	42.4	50.3	57.9	65.4	35.7
Oracle per-query α (bound)	32.1	41.3	43.9	45.9	47.1	32.1	39.6	41.0	41.8	42.1	32.1	16.6	11.3	6.3	1.4	31.2	47.9	54.1	60.1	65.4	41.4

Table 11: MS MARCO (A): all test metrics ($\times$100, mean over seeds), min–max normaliser unless stated.

	nDCG@k					MRR@k					P@k					R@k					
	1	3	5	10	50	1	3	5	10	50	1	3	5	10	50	1	3	5	10	50	MAP
BM25 (first stage / in-pool)	10.3	17.4	19.9	23.0	27.6	10.3	15.8	17.2	18.5	19.6	10.3	7.7	5.9	3.9	1.2	10.1	22.5	28.7	37.9	58.1	19.4
SBERT (s_{bi})	20.7	30.2	33.1	35.7	38.6	20.7	28.1	29.7	30.9	31.5	20.7	12.7	9.1	5.4	1.4	20.3	37.0	44.0	51.9	64.2	31.1
CE only (α=1)	25.4	36.2	39.1	41.5	43.3	25.4	33.9	35.6	36.6	37.0	25.4	15.2	10.6	6.1	1.4	24.8	43.9	50.9	58.1	65.3	36.5
GAT only (α=0)	19.9	30.1	33.0	35.7	38.4	19.9	27.9	29.4	30.6	31.2	19.9	12.9	9.2	5.5	1.4	19.5	37.6	44.6	52.7	64.6	30.7
ReGAT-BERT-style (a, b; BCE)	24.5	34.3	37.9	40.2	42.2	24.5	32.2	34.2	35.2	35.7	24.5	14.3	10.4	6.0	1.4	23.9	41.5	50.0	57.1	65.2	35.2
Fixed α (grid on val)	26.0	36.3	39.3	41.7	43.5	26.0	34.2	35.9	36.8	37.3	26.0	15.2	10.6	6.1	1.4	25.4	43.8	50.9	58.0	65.3	36.7
Fixed α (learned)	26.0	36.3	39.3	41.7	43.5	26.0	34.2	35.9	36.8	37.3	26.0	15.2	10.6	6.1	1.4	25.4	43.8	50.9	58.0	65.3	36.7
Adaptive $\alpha(q)$ Q	26.0	36.3	39.3	41.7	43.5	26.0	34.2	35.8	36.8	37.2	26.0	15.2	10.6	6.1	1.4	25.3	43.8	50.9	58.1	65.3	36.7
Adaptive $\alpha(q, s)$ QS	26.0	36.3	39.4	41.7	43.5	26.0	34.2	35.9	36.8	37.3	26.0	15.1	10.6	6.1	1.4	25.4	43.8	50.9	58.0	65.3	36.7
Adaptive $\alpha(s)$ S	26.1	36.3	39.4	41.7	43.5	26.1	34.2	35.9	36.9	37.3	26.1	15.1	10.6	6.1	1.4	25.4	43.8	50.9	58.0	65.3	36.7
Q, $\lambda = 0$	26.0	36.3	39.3	41.7	43.5	26.0	34.2	35.8	36.8	37.2	26.0	15.2	10.6	6.1	1.4	25.3	43.8	50.9	58.1	65.3	36.7
QS, $\lambda = 0$	26.0	36.3	39.4	41.7	43.5	26.0	34.2	35.9	36.8	37.3	26.0	15.1	10.6	6.1	1.4	25.4	43.8	50.9	58.0	65.3	36.7
S, $\lambda = 0$	26.1	36.3	39.4	41.7	43.5	26.1	34.2	35.9	36.9	37.3	26.1	15.1	10.6	6.1	1.4	25.4	43.8	50.9	58.0	65.3	36.7
Fixed α, BCE loss (zscore)	25.8	36.4	39.3	41.5	43.4	25.8	34.1	35.7	36.7	37.1	25.8	15.2	10.6	6.0	1.4	25.2	44.0	51.0	57.8	65.3	36.6
QS, BCE loss (zscore)	25.9	36.3	39.3	41.6	43.4	25.9	34.1	35.8	36.7	37.1	25.9	15.2	10.6	6.0	1.4	25.3	43.9	51.0	57.8	65.3	36.6
Oracle per-query α (bound)	32.0	41.8	44.3	46.0	47.3	32.0	39.9	41.2	41.9	42.2	32.0	16.9	11.5	6.3	1.4	31.1	48.8	54.9	60.1	65.4	41.5

Table 12: MS MARCO (B): all test metrics ($\times$100, mean over seeds), min–max normaliser unless stated.

	nDCG@k					MRR@k					P@k					R@k					
	1	3	5	10	50	1	3	5	10	50	1	3	5	10	50	1	3	5	10	50	MAP
BM25 (first stage / in-pool)	66.9	48.6	50.9	55.1	68.6	66.9	73.7	74.8	75.7	76.1	66.9	35.5	24.7	14.9	5.3	27.3	42.3	48.6	58.2	100.0	48.9
SBERT (s_{bi})	72.6	54.3	59.4	67.1	74.8	72.6	80.0	81.4	81.8	81.9	72.6	40.7	30.9	20.2	5.3	29.3	47.9	59.9	77.4	100.0	57.2
CE only (α=1)	79.8	61.9	65.4	71.1	78.7	79.8	86.3	87.0	87.3	87.4	79.8	46.5	32.9	20.1	5.3	32.4	55.3	64.3	77.4	100.0	62.6
GAT only (α=0)	72.4	53.9	58.9	66.8	74.7	72.4	79.5	80.9	81.5	81.6	72.4	40.2	30.6	20.2	5.3	29.3	47.4	59.1	77.1	100.0	57.1
ReGAT-BERT-style (a, b; BCE)	79.8	61.7	65.4	71.7	79.0	79.8	86.2	86.9	87.3	87.3	79.8	46.4	33.2	20.6	5.3	32.5	54.7	64.5	78.8	100.0	63.0
Fixed α (grid on val)	78.4	60.6	64.7	71.3	78.4	78.4	85.1	85.9	86.3	86.3	78.4	45.7	33.1	20.7	5.3	31.9	53.8	64.2	79.2	100.0	62.3
Fixed α (learned)	78.4	60.6	64.7	71.3	78.4	78.4	85.1	85.9	86.3	86.3	78.4	45.7	33.1	20.7	5.3	31.9	53.8	64.2	79.2	100.0	62.3
Adaptive $\alpha(q)$ Q	78.4	60.6	64.7	71.3	78.4	78.4	85.1	85.9	86.3	86.3	78.4	45.7	33.1	20.7	5.3	31.9	53.8	64.2	79.2	100.0	62.3
Adaptive $\alpha(q, s)$ QS	78.3	60.6	64.7	71.3	78.4	78.3	85.1	85.9	86.3	86.3	78.3	45.7	33.1	20.7	5.3	31.9	53.8	64.2	79.2	100.0	62.3
Adaptive $\alpha(s)$ S	78.4	60.6	64.7	71.3	78.4	78.4	85.1	85.9	86.3	86.3	78.4	45.7	33.1	20.7	5.3	31.9	53.8	64.2	79.2	100.0	62.3
Adaptive QS+C	78.4	60.6	64.7	71.3	78.4	78.4	85.1	85.9	86.3	86.3	78.4	45.7	33.1	20.7	5.3	31.9	53.8	64.2	79.2	100.0	62.3
Q, $\lambda = 0$	78.4	60.6	64.7	71.3	78.4	78.4	85.1	85.9	86.3	86.3	78.4	45.7	33.1	20.7	5.3	31.9	53.8	64.2	79.2	100.0	62.3
QS, $\lambda = 0$	78.3	60.6	64.7	71.3	78.4	78.3	85.1	85.9	86.3	86.3	78.3	45.7	33.1	20.7	5.3	31.9	53.8	64.2	79.2	100.0	62.3
S, $\lambda = 0$	78.4	60.6	64.7	71.3	78.4	78.4	85.1	85.9	86.3	86.3	78.4	45.7	33.1	20.7	5.3	31.9	53.8	64.2	79.2	100.0	62.3
Fixed α, BCE loss (zscore)	79.3	61.0	65.0	71.5	78.6	79.3	85.7	86.5	86.9	86.9	79.3	45.9	33.1	20.6	5.3	32.2	54.1	64.3	79.0	100.0	62.6
QS, BCE loss (zscore)	80.3	61.4	65.2	71.8	78.9	80.3	86.2	87.0	87.4	87.4	80.3	46.1	33.0	20.7	5.3	32.6	54.5	64.2	79.3	100.0	62.9
Oracle per-query α (bound)	87.9	68.5	72.0	78.7	83.5	87.9	91.9	92.3	92.5	92.5	87.9	51.8	36.4	22.5	5.3	35.6	61.1	70.7	85.8	100.0	69.7

Table 13: MuSiQue: all test metrics ($\times$100, mean over seeds), min–max normaliser unless stated. Closed pools: every gold passage is in the pool, so R@50 $= 1$ and P@50 = #gold/50 for all rows (also R@10 $= 1$ when $K = 10$).

	nDCG@k					MRR@k					P@k					R@k					
	1	3	5	10	50	1	3	5	10	50	1	3	5	10	50	1	3	5	10	50	MAP
BM25 (first stage / in-pool)	81.2	67.6	71.9	81.7	81.7	81.2	86.6	87.4	87.9	87.9	81.2	43.5	29.4	20.0	4.0	40.6	65.2	73.6	100.0	100.0	69.3
SBERT (s_{bi})	80.1	69.2	74.7	82.9	82.9	80.1	85.8	86.7	87.2	87.2	80.1	45.2	31.5	20.0	4.0	40.0	67.8	78.8	100.0	100.0	71.7
CE only (α=1)	92.2	81.0	84.7	89.8	89.8	92.2	95.1	95.3	95.4	95.4	92.2	52.9	34.7	20.0	4.0	46.1	79.4	86.7	100.0	100.0	81.8
GAT only (α=0)	81.7	71.1	76.7	84.1	84.1	81.7	87.2	88.0	88.4	88.4	81.7	46.6	32.5	20.0	4.0	40.8	69.9	81.1	100.0	100.0	73.5
ReGAT-BERT-style (a, b; BCE)	92.1	81.1	84.8	89.9	89.9	92.1	95.0	95.3	95.3	95.3	92.1	52.9	34.7	20.0	4.0	46.1	79.4	86.7	100.0	100.0	81.9
Fixed α (grid on val)	92.0	81.1	84.8	89.9	89.9	92.0	95.0	95.2	95.3	95.3	92.0	53.0	34.7	20.0	4.0	46.0	79.5	86.8	100.0	100.0	81.9
Fixed α (learned)	92.1	81.1	84.8	89.9	89.9	92.1	95.0	95.2	95.3	95.3	92.1	53.0	34.7	20.0	4.0	46.0	79.5	86.7	100.0	100.0	82.0
Adaptive $\alpha(q)$ Q	92.0	81.1	84.8	89.9	89.9	92.0	95.0	95.2	95.3	95.3	92.0	53.0	34.7	20.0	4.0	46.0	79.5	86.8	100.0	100.0	81.9
Adaptive $\alpha(q, s)$ QS	92.3	81.1	84.8	89.9	89.9	92.3	95.2	95.4	95.5	95.5	92.3	53.0	34.7	20.0	4.0	46.2	79.4	86.8	100.0	100.0	81.9
Adaptive $\alpha(s)$ S	92.3	81.2	84.8	89.9	89.9	92.3	95.2	95.4	95.5	95.5	92.3	53.0	34.7	20.0	4.0	46.2	79.5	86.8	100.0	100.0	82.0
Adaptive QS+C	92.3	81.1	84.8	89.9	89.9	92.3	95.2	95.5	95.5	95.5	92.3	53.0	34.7	20.0	4.0	46.2	79.4	86.8	100.0	100.0	81.9
Q, $\lambda = 0$	92.0	81.1	84.8	89.9	89.9	92.0	95.0	95.2	95.3	95.3	92.0	53.0	34.7	20.0	4.0	46.0	79.5	86.8	100.0	100.0	81.9
QS, $\lambda = 0$	92.4	81.1	84.9	89.9	89.9	92.4	95.2	95.5	95.5	95.5	92.4	53.0	34.7	20.0	4.0	46.2	79.5	86.8	100.0	100.0	81.9
S, $\lambda = 0$	92.3	81.1	84.8	89.9	89.9	92.3	95.2	95.4	95.5	95.5	92.3	53.0	34.7	20.0	4.0	46.1	79.5	86.8	100.0	100.0	81.9
Fixed α, BCE loss (none)	91.9	81.0	84.8	89.8	89.8	91.9	94.9	95.2	95.2	95.2	91.9	53.0	34.8	20.0	4.0	46.0	79.4	86.9	100.0	100.0	81.9
QS, BCE loss (none)	91.9	81.0	84.8	89.9	89.9	91.9	94.9	95.2	95.3	95.3	91.9	53.0	34.8	20.0	4.0	46.0	79.4	86.9	100.0	100.0	81.9
Oracle per-query α (bound)	96.0	86.1	89.4	92.9	92.9	96.0	97.5	97.7	97.7	97.7	96.0	56.4	36.5	20.0	4.0	48.0	84.5	91.1	100.0	100.0	86.8

Table 14: HotpotQA: all test metrics ($\times$100, mean over seeds), min–max normaliser unless stated. Closed pools: every gold passage is in the pool, so R@50 $= 1$ and P@50 = #gold/50 for all rows (also R@10 $= 1$ when $K = 10$).

	nDCG@k					MRR@k					P@k					R@k					
	1	3	5	10	50	1	3	5	10	50	1	3	5	10	50	1	3	5	10	50	MAP
BM25 (first stage / in-pool)	73.8	58.4	61.1	63.3	66.2	73.8	79.1	79.9	80.3	80.6	73.8	36.6	24.1	13.2	3.0	36.9	54.9	60.3	65.8	76.3	55.0
SBERT (s_{bi})	62.7	48.4	51.1	53.7	57.8	62.7	68.6	69.7	70.3	70.8	62.7	30.1	20.2	11.4	2.9	31.4	45.1	50.4	57.0	72.2	45.4
CE only (α=1)	84.6	68.6	71.0	72.6	74.0	84.6	88.4	88.8	89.0	89.1	84.6	43.4	27.9	14.8	3.1	42.3	65.1	69.8	73.9	78.8	64.8
GAT only (α=0)	65.7	50.6	53.0	55.5	59.4	65.7	71.2	72.0	72.6	73.0	65.7	31.4	20.7	11.6	2.9	32.9	47.1	51.8	58.2	72.6	47.3
ReGAT-BERT-style (a, b; BCE)	79.9	64.1	66.7	68.8	70.9	79.9	84.5	85.1	85.4	85.5	79.9	40.4	26.3	14.2	3.1	40.0	60.6	65.8	71.0	78.2	60.6
Fixed α (grid on val)	84.6	68.6	71.0	72.6	74.1	84.6	88.4	88.8	89.0	89.1	84.6	43.4	27.9	14.8	3.1	42.3	65.1	69.8	73.8	78.8	64.9
Fixed α (learned)	84.6	68.6	71.0	72.6	74.1	84.6	88.4	88.8	89.0	89.1	84.6	43.4	27.9	14.8	3.1	42.3	65.1	69.8	73.8	78.8	64.9
Adaptive $\alpha(q)$ Q	84.6	68.6	71.0	72.6	74.1	84.6	88.4	88.8	89.0	89.1	84.6	43.4	27.9	14.8	3.1	42.3	65.1	69.8	73.8	78.8	64.9
Adaptive $\alpha(q, s)$ QS	84.6	68.6	71.0	72.6	74.1	84.6	88.4	88.8	89.0	89.1	84.6	43.4	27.9	14.8	3.1	42.3	65.1	69.8	73.8	78.8	64.9
Adaptive $\alpha(s)$ S	84.6	68.6	71.0	72.6	74.1	84.6	88.4	88.8	89.0	89.1	84.6	43.4	27.9	14.8	3.1	42.3	65.1	69.8	73.8	78.8	64.9
Adaptive QS+C	84.6	68.6	71.0	72.6	74.1	84.6	88.4	88.8	89.0	89.1	84.6	43.4	27.9	14.8	3.1	42.3	65.1	69.8	73.8	78.8	64.9
Q, $\lambda = 0$	84.6	68.6	71.0	72.6	74.1	84.6	88.4	88.8	89.0	89.1	84.6	43.4	27.9	14.8	3.1	42.3	65.1	69.8	73.8	78.8	64.9
QS, $\lambda = 0$	84.6	68.6	71.0	72.6	74.1	84.6	88.4	88.8	89.0	89.1	84.6	43.4	27.9	14.8	3.1	42.3	65.1	69.8	73.8	78.8	64.9
S, $\lambda = 0$	84.6	68.6	71.0	72.6	74.1	84.6	88.4	88.8	89.0	89.1	84.6	43.4	27.9	14.8	3.1	42.3	65.1	69.8	73.8	78.8	64.9
Fixed α, BCE loss (none)	84.8	68.7	71.1	72.7	74.1	84.8	88.5	89.0	89.1	89.2	84.8	43.4	28.0	14.8	3.1	42.4	65.1	69.9	73.9	78.8	64.9
QS, BCE loss (none)	84.8	68.7	71.1	72.7	74.1	84.8	88.5	89.0	89.1	89.2	84.8	43.4	28.0	14.8	3.1	42.4	65.1	69.9	73.9	78.8	64.9
Oracle per-query α (bound)	89.3	72.3	74.3	75.6	76.5	89.3	91.6	91.9	92.0	92.1	89.3	45.5	28.8	15.1	3.1	44.6	68.3	72.1	75.3	78.8	68.4

Table 15: BEIR-HotpotQA: all test metrics ($\times$100, mean over seeds), min–max normaliser unless stated.

	nDCG@k					MRR@k					P@k					R@k					
	1	3	5	10	50	1	3	5	10	50	1	3	5	10	50	1	3	5	10	50	MAP
BM25 (first stage / in-pool)	93.8	75.9	78.1	87.8	87.8	93.8	96.4	96.5	96.5	96.5	93.8	52.7	36.1	24.4	4.9	41.6	67.9	77.1	100.0	100.0	76.5
SBERT (s_{bi})	94.4	78.1	80.9	89.2	89.2	94.4	96.8	96.9	96.9	96.9	94.4	54.8	38.3	24.4	4.9	41.9	70.5	80.7	100.0	100.0	79.1
CE only (α=1)	96.6	81.1	83.0	90.8	90.8	96.6	98.1	98.2	98.2	98.2	96.6	56.8	38.6	24.4	4.9	42.9	73.4	82.3	100.0	100.0	81.6
GAT only (α=0)	95.5	81.6	84.7	91.0	91.0	95.5	97.5	97.5	97.5	97.5	95.5	58.1	40.8	24.4	4.9	42.5	74.5	85.5	100.0	100.0	82.4
ReGAT-BERT-style (a, b; BCE)	97.3	83.0	85.0	91.7	91.7	97.3	98.5	98.6	98.6	98.6	97.3	58.6	40.1	24.4	4.9	43.3	75.5	84.7	100.0	100.0	83.4
Fixed α (grid on val)	97.3	82.9	85.2	91.8	91.8	97.3	98.5	98.5	98.5	98.5	97.3	58.6	40.3	24.4	4.9	43.3	75.5	85.0	100.0	100.0	83.4
Fixed α (learned)	97.3	82.9	85.2	91.8	91.8	97.3	98.5	98.5	98.5	98.5	97.3	58.6	40.3	24.4	4.9	43.3	75.5	85.0	100.0	100.0	83.4
Adaptive $\alpha(q)$ Q	97.0	83.0	85.4	91.8	91.8	97.0	98.3	98.4	98.4	98.4	97.0	58.8	40.5	24.4	4.9	43.1	75.7	85.4	100.0	100.0	83.6
Adaptive $\alpha(q, s)$ QS	97.0	82.6	85.0	91.6	91.6	97.0	98.4	98.4	98.4	98.4	97.0	58.3	40.2	24.4	4.9	43.2	75.2	84.9	100.0	100.0	83.3
Adaptive $\alpha(s)$ S	97.3	82.9	85.2	91.8	91.8	97.3	98.5	98.5	98.5	98.5	97.3	58.6	40.3	24.4	4.9	43.3	75.5	85.0	100.0	100.0	83.4
Adaptive QS+C	97.0	82.7	85.1	91.6	91.6	97.0	98.3	98.4	98.4	98.4	97.0	58.4	40.3	24.4	4.9	43.1	75.3	85.0	100.0	100.0	83.3
Q, $\lambda = 0$	96.9	83.0	85.4	91.8	91.8	96.9	98.3	98.3	98.3	98.3	96.9	58.8	40.6	24.4	4.9	43.1	75.7	85.5	100.0	100.0	83.6
QS, $\lambda = 0$	97.3	82.9	85.2	91.8	91.8	97.3	98.5	98.5	98.5	98.5	97.3	58.6	40.3	24.4	4.9	43.3	75.5	85.0	100.0	100.0	83.4
S, $\lambda = 0$	97.3	82.9	85.2	91.8	91.8	97.3	98.5	98.5	98.5	98.5	97.3	58.6	40.3	24.4	4.9	43.3	75.5	85.0	100.0	100.0	83.4
Fixed α, BCE loss (rank)	96.6	82.0	84.4	91.3	91.3	96.6	98.2	98.2	98.2	98.2	96.6	57.7	39.8	24.4	4.9	43.0	74.6	84.3	100.0	100.0	82.6
QS, BCE loss (rank)	96.5	82.9	85.7	91.7	91.7	96.5	98.1	98.1	98.1	98.1	96.5	59.0	41.1	24.4	4.9	42.9	75.8	86.2	100.0	100.0	83.6
Oracle per-query α (bound)	99.0	87.3	89.3	94.0	94.0	99.0	99.5	99.5	99.5	99.5	99.0	62.5	42.6	24.4	4.9	44.1	80.3	89.5	100.0	100.0	87.6

Table 16: 2Wiki: all test metrics ($\times$100, mean over seeds), min–max normaliser unless stated. Closed pools: every gold passage is in the pool, so R@50 $= 1$ and P@50 = #gold/50 for all rows (also R@10 $= 1$ when $K = 10$).

		TREC DL 2019			TREC DL 2020		
		nDCG@10	MRR@10	MAP	nDCG@10	MRR@10	MAP
MS MARCO (A)	BM25 (first stage / in-pool)	50.6	70.2	24.8	48.0	65.3	26.9
	SBERT (s_{bi})	66.9	80.9	31.1	60.9	74.5	35.1
	CE only (α=1)	72.7	87.5	35.7	67.6	84.6	39.6
	GAT only (α=0)	67.3	80.4	31.4	59.4	71.8	34.5
	ReGAT-BERT-style (a, b; BCE)	72.2	85.3	34.7	65.3	83.0	38.7
	Fixed α (grid on val)	72.2	86.1	35.3	67.1	83.9	39.2
	Fixed α (learned)	72.2	86.1	35.3	67.1	83.9	39.2
	Adaptive $\alpha(q)$ Q	72.2	86.1	35.3	67.1	83.9	39.2
	Adaptive $\alpha(q, s)$ QS	72.2	85.9	35.3	67.1	83.9	39.2
	Adaptive $\alpha(s)$ S	72.3	86.1	35.3	67.1	83.9	39.2
	Oracle per-query α (bound)	74.8	89.5	35.8	69.9	86.4	40.5
MS MARCO (B)	BM25 (first stage / in-pool)	50.6	70.2	24.8	48.0	65.3	26.9
	SBERT (s_{bi})	66.9	80.9	31.1	60.9	74.5	35.1
	CE only (α=1)	72.7	87.5	35.7	67.6	84.6	39.6
	GAT only (α=0)	66.0	79.5	31.0	59.8	75.7	35.1
	ReGAT-BERT-style (a, b; BCE)	71.7	84.9	34.8	65.3	83.7	38.5
	Fixed α (grid on val)	72.3	87.1	35.7	67.2	83.9	39.6
	Fixed α (learned)	72.3	87.1	35.7	67.2	83.9	39.6
	Adaptive $\alpha(q)$ Q	72.5	87.1	35.7	67.2	83.9	39.5
	Adaptive $\alpha(q, s)$ QS	72.5	87.1	35.7	67.2	83.9	39.6
	Adaptive $\alpha(s)$ S	72.4	87.1	35.7	67.2	83.8	39.5
	Oracle per-query α (bound)	74.8	89.5	35.9	70.2	87.2	40.5

Table 17: Out-of-distribution TREC DL (graded nDCG; MRR and MAP with rel $\geq$ 2), $\times$100. Not used for any decision.

			Test nDCG@10		Best fixed α in bucket		% queries	Learned
Dataset	Bucket	n (test)	CE	GAT	val	test [95% CI]	prefer ctx	$\bar{\alpha}$ (gate)
MS MARCO (A)	short ($\leq$4 w)	2,149	45.0	40.3	0.70	0.95 [0.70, 0.95]	35	–
	medium (5–7 w)	3,291	41.6	35.3	0.45	1.00 [0.85, 1.00]	33	–
	long ($\geq$8 w)	1,540	35.8	28.8	0.75	0.95 [0.80, 1.00]	31	–
MuSiQue	2-hop	1,252	73.8	68.0	0.40	0.80 [0.65, 0.80]	40	0.40
	3-hop	760	71.8	66.7	0.40	0.55 [0.55, 0.85]	39	0.40
	4-hop	405	61.3	63.3	0.25	0.40 [0.25, 0.55]	52	0.40
HotpotQA	comparison	1,487	95.6	94.4	0.40	0.55 [0.45, 0.70]	40	0.90
	bridge	5,918	88.4	81.6	1.00	0.95 [0.90, 1.00]	33	0.88
BEIR-HotpotQA	comparison	1,487	85.7	77.2	0.65	0.70 [0.65, 0.75]	27	0.95
	bridge	5,918	69.3	50.0	1.00	1.00 [1.00, 1.00]	12	0.95
2Wiki	comparison	3,040	97.3	95.5	0.90	0.65 [0.65, 0.80]	30	0.72
	bridge-comp.	2,751	90.1	92.0	0.25	0.20 [0.00, 0.20]	67	0.35
	compositional	5,236	88.5	88.9	0.45	0.50 [0.35, 0.50]	53	0.45
	inference	1,549	86.7	87.6	0.25	0.45 [0.25, 0.60]	59	0.30

Table 18: Query-level test of the hypothesis (min–max). Best fixed α = arg-max of the bucket's mean nDCG@10 curve; CI from 2,000 bootstrap resamples of the bucket's queries. "% prefer ctx": among queries whose nDCG@10 changes with α, the share whose optimal α-interval is centred below 0.5 (tie-robust; the remaining queries are indifferent). Learned $\bar{\alpha}$: mean α of the best (on val) in-domain gate in the bucket. Analysis only: no label enters any gate.

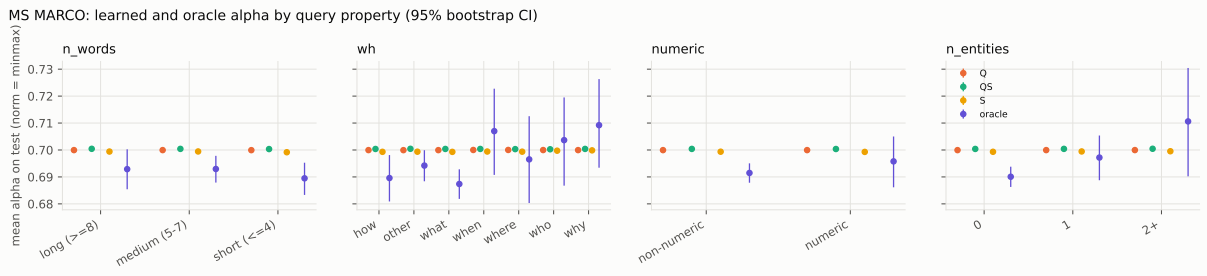


Figure 8: MS MARCO (A, test, min–max): mean learned α of the three gates and mean oracle α by query property, with 95% bootstrap CIs.

	val			test		
Graph	2	3	4	2	3	4
none (self-loops)	**0.35**	**0.25**	**0.15**	0.80	**0.45**	**0.25**
random	**0.45**	0.50	**0.25**	0.65	0.50	**0.45**
kNN emb. (default)	**0.40**	**0.40**	**0.25**	0.80	0.55	**0.40**
title link	**0.35**	**0.25**	**0.15**	0.75	0.55	**0.35**
kNN ∪ title	**0.40**	**0.25**	**0.15**	0.75	0.65	**0.30**

Table 19: MuSiQue: best fixed α (min–max) by hop count for a GAT retrained on each graph type. The 4-hop optimum is the lowest and below 0.5 (bold) for every graph.

Dataset	Graph	Degree	Gold pairs linked (%)	GAT nDCG@10	α_0	Fixed nDCG@10	Δ adaptive (best gate)	Headroom (oracle)
MuSiQue	none (self-loops)	0.00	0	67.7	0.35	71.75	-0.03	7.7
	random (same degrees)	2.78	14	67.0	0.50	71.89	-0.03	6.6
	kNN emb. (default)	2.78	27	66.8	0.40	71.28	+0.00	7.4
	title link	0.81	19	68.0	0.35	71.97	+0.00	7.6
	kNN $\cup$ title	3.20	36	67.0	0.40	71.40	+0.00	7.4
HotpotQA	none (self-loops)	0.00	0	84.6	0.85	89.98	+0.02	3.0
	random (same degrees)	2.78	31	83.7	0.90	89.89	-0.02	2.8
	kNN emb. (default)	2.78	48	84.1	0.85	89.87	+0.02	3.0
	title link	0.98	68	87.4	0.65	90.52	+0.05	3.5
	kNN $\cup$ title	3.18	80	85.5	0.65	89.97	+0.18$^{\blacktriangle}$	3.4

Table 20: Graph controls (H4, exploratory): a GAT retrained per graph type, everything else fixed. Degree excludes self-loops; "gold pairs linked" = share of supporting-paragraph pairs joined by an edge (test). nDCG@10 and Δ are $\times 100$ on test; Δ = best gate (on val) $-$ best fixed α, $^{\blacktriangle}$ $p < 0.05$.

Data	Graph	Deg.	GAT	s_{bi}
MS MARCO (A)	no_graph	1.0	37.88	37.71
MS MARCO (A)	random_graph	12.3	37.94	37.71
MS MARCO (A)	knn_k4_s0.0	7.1	37.73	37.71
MS MARCO (A)	knn_k4_s0.3	6.9	37.76	37.71
MS MARCO (A)	knn_k4_s0.5	5.6	37.65	37.71
MS MARCO (A)	knn_k8_s0.0	13.2	38.00	37.71
MS MARCO (A)	knn_k8_s0.3	12.3	37.98	37.71
MS MARCO (A)	knn_k8_s0.5	9.2	37.90	37.71
MS MARCO (A)	knn_k16_s0.0	24.6	38.04	37.71
MS MARCO (A)	knn_k16_s0.3	22.0	37.92	37.71
MS MARCO (A)	knn_k16_s0.5	14.7	37.76	37.71
MuSiQue	knn_k2	2.0	75.62	70.08
MuSiQue	knn_k3	2.6	75.19	70.08
MuSiQue	knn_k5	3.4	75.27	70.08
HotpotQA	knn_k2	2.2	84.26	82.67
HotpotQA	knn_k3	2.8	84.09	82.67
HotpotQA	knn_k5	3.5	83.93	82.67

Table 21: GAT graph sensitivity on val (nDCG@10 $\times 100$ of s_{ctx} alone). knn_kk_sm: k neighbours, min cosine m. MS MARCO degrees include the self-loop.

Hyp.	Dataset	Gate	Statistic	Value	95% CI	p_{Holm}	Pred.	Verdict
H1	MuSiQue	best=Q	Spearman(α, hops)	+0.107	[+0.067, +0.147]	0.001	<0	contradicted
H2	MuSiQue	best=Q	gain − gain_{MSM}	-0.004	[-0.025, +0.017]	1.000	>0	inconclusive
H3	MuSiQue	oracle	headroom − $\text{headroom}_{\text{MSM}}$	+2.435	[+1.942, +2.932]	0.001	>0	supported
H1	HotpotQA	best=QS	$\bar{\alpha}_{\text{bridge}} - \bar{\alpha}_{\text{comp.}}$	-0.026	[-0.037, -0.016]	0.001	<0	supported
H2	HotpotQA	best=QS	gain − gain_{MSM}	+0.011	[-0.095, +0.117]	1.000	>0	inconclusive
H3	HotpotQA	oracle	headroom − $\text{headroom}_{\text{MSM}}$	-2.036	[-2.365, -1.710]	0.001	>0	contradicted
H1	BEIR-HotpotQA	best=Q	$\bar{\alpha}_{\text{bridge}} - \bar{\alpha}_{\text{comp.}}$	+0.000	[+0.000, +0.000]	0.001	<0	contradicted
H2	BEIR-HotpotQA	best=Q	gain − gain_{MSM}	-0.005	[-0.025, +0.016]	1.000	>0	inconclusive
H3	BEIR-HotpotQA	oracle	headroom − $\text{headroom}_{\text{MSM}}$	-2.078	[-2.427, -1.739]	0.001	>0	contradicted

Table 22: Pre-registered primary tests (written before any multi-hop run; Holm over the nine tests). H1 uses the best (on val) gate; H2/H3 values are nDCG@10 $\times$100. H1 verdicts on MuSiQue and BEIR are computed on gates whose α varies by less than 10^{-3} across queries (Table 8), so their sign is not interpretable.